Exercise - 6.2

1. Find the equation of the lines passing through the point (1,1)
   (i) with y-intercept (-4)
   (ii) with slope 3
   (iii) and (-2, 3)
   (iv) and the perpendicular from the origin makes an angle $60°$ with x- axis.
2. If $P(r, c)$ is mid point of a line segment between the axes, then show that $\frac{x}{r} + \frac{y}{c} = 2$.
3. Find the equation of the line passing through the point $(1, 5)$ and also divides the co-ordinate axes in the ratio 3:10.
4. If p is length of perpendicular from origin to the line whose intercepts on the axes are a and b, then show that $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$.
5. The normal boiling point of water is $100°C$ or $212°F$ and the freezing point of water is $0°C$ or $32°F$. (i) Find the linear relationship between C and F
   Find (ii) the value of C for $98.6°F$ and (iii) the value of F for $38°C$
6. An object was launched from a place P in constant speed to hit a target. At the 15th second it was $1400\,m$ away from the target and at the 18th second 800m away. Find (i) the distance between the place and the target (ii) the distance covered by it in 15 seconds.(iii) time taken to hit the target.
7. Population of a city in the years 2005 and 2010 are 1,35,000 and 1,45,000 respectively.
   Find the approximate population in the year 2015. (assuming that the growth of population is constant)
8. Find the equation of the line, if the perpendicular drawn from the origin makes an angle $30°$ with x-axis and its length is 12.
9. Find the equation of the straight lines passing through (8, 3) and having intercepts whose sum is 1
10. Show that the points (1, 3), (2, 1) and $\left(\frac{1}{2}, 4\right)$ are collinear, by using (i) concept of slope (ii) using a straight line and (iii) any other method
11. A straight line is passing through the point $A(1, 2)$ with slope $\frac{5}{12}$. Find points on the line which are 13 units away from A.
12. A 150 m long train is moving with constant velocity of 12.5 m/s. Find (i) the equation of the motion of the train, (ii) time taken to cross a pole. (iii) The time taken to cross the bridge of length 850 m is?
13. A spring was hung from a hook in the ceiling. A number of different weights were attached to the spring to make it stretch, and the total length of the spring was measured each time is shown in the following table.

Weight (kg)	2	4	5	8
Length (cm)	3	4	4.5	6

   (i) Draw a graph showing the results.
   (ii) Find the equation relating the length of the spring to the weight on it.
   (iii) What is the actual length of the spring.
   (iv) If the spring has to stretch to 9 cm long, how much weight should be added?
   (v) How long will the spring be when 6 kilograms of weight on it?